```
this.children.add(newChild2)
          #12

node.updateBoundingEnvelope()
          #13
      if this.children.size <= M
then                                                  #14
        return null
  return
this.split()
      #15
```

#1 Method `insert` takes a node and a point, and adds the point to the node's subtree. It is defined recursively, and returns `null` if `node` doesn't need to be split as a result of the insertion, otherwise it returns the pair of nodes resulting from splitting `node`.
#2 Check if `node` is a leaf.
#3 If it is a leaf, check if already contains the argument among its points, and if it does, we can return. #4 Otherwise, add the point to the leaf.
#5 We need to re-compute the centroid and radius for this leaf, after adding the new point.
#6 If we added a new point, we need to check if now this leaf holds more than `M` points: if there are no more than `M`, we can return, otherwise we continue to line #13.
#7 If we are in an internal node, we need to find which branch to traverse, calling a helper method.
#8 Recursively traverse the tree and insert the new point, storing the outcome of the operation.